



Content and quantity of highlights and annotations predict learning from multiple digital texts

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ARTICLE INFO

Keywords:

Digital Reading
Highlights
Annotations
Reading Strategies
Integration
Metacognition

ABSTRACT

Learning from multiple texts is an essential yet challenging academic task. Thus, a persistent need in the field is identifying those strategies that will support students' learning from multiple texts, and facilitating their use. In this study, we adopt a quasi-experimental design to examine how the use of digital annotations, allowing undergraduates ($N = 278$) to identify and, potentially, link important concepts across texts, can be used to foster learning from multiple texts. Additionally, we examine students' digital highlights and annotations under four different strategy conditions, assigned to elicit different forms of processing found to be effective in prior work. These conditions included asking students to highlight and annotate (a) important or relevant information in texts, (b) connections or links across texts (i.e., intertextual reasoning condition), (c) information in texts facilitating evaluation, and (d) difficult to understand information (i.e., metacognitive monitoring condition). Some differences were found in students' rendering of highlights and annotations across strategy conditions, however, no differences in multiple text comprehension and integration performance were found across strategy conditions. Still, membership in the metacognitive monitoring strategy condition, relative to the relevance condition, as well as students' highlighting of information supporting evaluation (i.e., source information, statistical evidence), and accessing of references during reading were predictive of multiple text comprehension and integration. Thus, this study contributes to theory by providing further evidence of strategies (e.g., metacognitive monitoring, evaluation) supportive of multiple text learning and introduces a promising means whereby such strategy use may be fostered.

1. Introduction

Being able to integrate information across disparate resources in order to make sense of complex topics and complete academic tasks has been recognized as a key competency for today's digital age learners (National Governors Association Center for Best Practices, 2010; OECD, 2017). It is also a formidable challenge requiring that learners manage and coordinate an immense volume of information across disparate sources, complementing or conflicting with one another, all while monitoring comprehension (List & Alexander, 2018; Afflerbach & Cho, 2009). Beyond cataloging the variety of challenges that students experience when learning from multiple texts, particularly online, studies have identified the strategies that students use to help them overcome such challenges and learn effectively (List, 2020; List & Alexander, 2020; List & Du, 2021; Schoor et al., 2021). Yet determining how to support

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effective strategy use remains a challenge (Barzilai et al., 2018). In this exploratory study, we take advantage of recent work suggesting the promise of digital annotations (Chiu et al., 2022; Sievers, 2021), to examine how these can support students' learning from multiple digital texts. Moreover, given findings on the role of task assignment in shaping learning (List et al., 2019; Schoor et al., 2021), we further examine whether directing students to engage in one of four different modes of processing can improve the quality of students' digital annotations and facilitate performance. In this way, we seek to contribute to the literature on learning from multiple texts by examining the extent to which multiple text learning can be supported via the use of digital annotations. Moreover, we contribute to the literature on digital annotations by examining whether the type of task instructions that students receive, prior to processing, can improve the quality of the annotations produced, resulting in increased performance. In particular, given the widespread availability of highlighting and annotation tools for digital reading (MacGregor & Salaz, 2019), students' use of such tools to support learning from multiple texts is examined in this study.

1.1. Models of learning from multiple texts

A number of models of learning from multiple texts have been proposed, among the most prominent of these is the Multiple Documents Task-Based Relevance Assessment and Content Extraction (MD-TRACE) model (Rouet & Britt, 2011). The MD-TRACE conceptualizes learning from multiple texts as a five-step process. In Step 1, students construct a *task model* or a cognitive representation of an externally assigned task. In Step 2, students determine that they have an *information need* or require multiple text access to meet task goals. In Step 3, learners select, process, and integrate multiple sources of information to produce an externalized task product (Step 4). Finally, students determine whether or not the product that they created satisfies task demands (Step 5) and potentially return to earlier steps of the model.

Step 3, including students' multiple text processing and integration, is the focus of the present investigation. In particular students' integration, or cognitive conceptualization of multiple texts, has been described in the Documents Model Framework (DMF, Britt et al., 1999; Perfetti et al., 1999), a framework complementary to MD-TRACE. The DMF proposes that students construct two types of cognitive representations when learning from multiple texts. The *integrated mental model* represents a coherent understanding of the common content or situation described across texts. The *intertext model* is structural, including students' linking of text-based information to the specific sources it originates from and formation of source-level connections (i.e., identifying texts as agreeing, disagreeing, or otherwise related to one another). Students can then link their integrated mental models and intertext models in various ways to cognitively conceptualize multiple texts, such that connections between specific content within texts and documents-level relations across texts are formed. Here, we examine the extent to which students are successful in integrating content across texts and whether such integration can be fostered through students' annotation of digital texts.

1.2. Strategies for learning from multiple texts

A variety of strategies have been associated with learning from multiple texts (List, 2020; Perret, 2021). Anmarkrud et al. (2014) conducted a think-aloud study, finding that students engaged three categories of strategic processing. These include students: (a) *identifying and learning important information* (e.g., connecting text-based information to content in other texts or to prior knowledge) (b) *monitoring* (e.g., identifying comprehension problems and seeking to resolve these), and (c) *evaluating* (e.g., using source information to interpret content in texts; corroborating information across texts). They determined that students carried out each of these processes in reference to either a single text or across multiple texts. Most commonly, students formed cross-textual links to identify and learn important information (47.1%) and to evaluate information (36.3%), rather than to monitor comprehension (16.7%). Total strategy use, as well as students' engagement in monitoring, evaluating, and linking strategies, were all associated with argumentative reasoning performance, as assessed via writing (Anmarkrud et al., 2014). Elsewhere, verbatim *paraphrasing* of information from texts, *self-explaining* or elaborating information from texts, based on prior knowledge, *monitoring comprehension* and *evaluating* have all been determined, via think-alouds, to be strategies engaged during learning. And engagement in self-explanation, in particular, has been associated with improved multiple text comprehension (Linderholm et al., 2014; McCrudden et al., 2022; Wolfe & Goldman, 2005).

Beyond think-alouds, students' notes, composed during multiple text use, have been examined as a fertile, and potentially less obtrusive and more naturalistic, means of capturing strategic processing (Kobayashi, 2009; Lee, 2021). For example, Hagen et al. (2014) examined students' notes, spontaneously recorded while students read seven texts about climate change. Information in notes was coded as reflecting (a) *paraphrases*, if it restated or transformed texts' content in a limited way, (b) *intratextual elaboration*, if it demonstrated connection formation within a text or between information in text and prior knowledge, or (c) *intertextual elaboration*, if it linked information across two or more texts. The amount of paraphrased information included in students' notes was negatively associated with the comprehension of individual texts included in the document set (i.e., as assessed via a sentence verification task and an intratextual integration task). Conversely, the number of intertextual elaborations included in students' notes was positively associated with both intratextual and intertextual integration performance. List & Du (2021) coded students' notes according to the number of *text-based sentences* these included, as well as whether these featured *cross-textual connections*, evidence of students' *prior knowledge or opinions* (e.g., "I agree"), *source information*, or evidence of *evaluation*. They found both the inclusion of source information in students' notes and the number of cross-textual connections formed to predict the number of text-based sentences included in students' writing. Moreover, the number of cross-textual connections included in students' notes significantly predicting writing quality, overall.

1.3. Highlights and annotations

Still, in all of these studies, students' notes were composed in a stand-alone fashion, separately from texts, rather than these being embedded within or alongside texts' content. Students' notes co-located with texts' content are referred to as *annotations* (Yeh, Hung, & Chiang, 2017). Here, we examine whether students' digital annotations, like separately composed notes, can support learning from multiple texts. Two broad types of annotations have been identified in prior work, with these differing in both function and form. Underlines, highlights, and other markings have been examined as emphatic, identifying important or otherwise notable information for students to return to following reading. Comments or brief notes have been examined as generative in nature, with summaries, interpretations, and other types of additions produced (Johnson et al., 2010; Marshall, 1997).

Studies examining learners' single text annotations has generally found these to be facilitative of comprehension (Azmuiddin et al., 2020; Clinton-Lisell et al., 2021; Zhu et al., 2020); although this has not always been the case (Inie et al., 2021; Li et al., 2016) and has been found to differ across print and digital mediums. Ben-Yehudah and Eshet-Alkalai (2014; 2018) compared students' print and digital reading, with a highlighting and annotation tool provided or not. In addition to finding an advantage for print reading, they found highlighting print (but not digital texts) to support students' performance on inference-level questions, but not factual level questions. Goodwin et al. (2020) found students to render significantly more highlights ($M = 4.39$, $SD = 8.62$) and annotations ($M = 0.40$, $SD = 1.43$) when reading on paper rather than when reading digitally (highlights: $M = 2.18$, $SD = 4.52$, annotations: $M = 0.08$, $SD = 0.43$). Further, unlike Ben-Yehudah and Eshet-Alkalai (2014; 2018), Goodwin et al. found highlight quantity to positively predict digital reading comprehension, while negatively predicting print reading comprehension. Students' annotation volume had no significant effect in either medium. Given these conflicting and still limited results, there is a need to examine the ways in which highlights and annotations can support students' text-based learning during digital reading.

Here, we explore how highlights and annotations can improve students' learning from, or the comprehension and integration of, multiple texts. Porter-O'Donnell (2004) suggests that textual annotation serves at least four purposes, including slowing down the pace of reading, making reading a more actively engaged and analytic process (e.g., increasing students' attendance to authors' use of language), and improving subsequent writing. The functionality of annotations in supporting students' more analytic reading is of particular interest in this study, as students' analytic attendance to specific terms or concepts within one text, may support their linking of concepts across texts and inter-textual corroboration (List et al., 2021; Kobayashi, 2009). That is, due to their insertion into or inclusion alongside texts' content, annotations may better support generative processing (e.g., cross-textual connection formation; Chen et al., 2020). Such generative processing may be considered preferable to students simply restating information from texts, as they have often been found to do when composing stand-alone notes (Du & List, 2020; List & Du, 2021; Hagen et al., 2014).

1.4. Highlights and annotations in multiple text learning

Investigations of annotations of multiple texts in digital contexts have been limited. In one recent study, Leroy and Kammerer (2022) asked students to read five partially-conflicting, digitally-presented texts, with a highlighting tool provided or not. They found that although the number of inter-textual connections that students formed in their essays did not differ between highlighting-tool provision conditions, students who reread more information for longer *and* had the highlighting tool available had significantly higher inter-textual integration performance. No differences in intertextual integration performance were found among students re-reading to a limited extent (i.e., regardless of whether they were provided with the highlighting tool or not). Leroy et al. (2021) examined the effects of both the availability of a text highlighting tool (or not) and the simultaneous presentation of multiple documents (rather than documents being presented sequentially) on integrated understanding. They, again, found a complex interaction. Both the simultaneous presentation of documents and the availability of the highlighting tool fostered intertextual understanding, but only under conditions wherein students' text revisits were frequent. Thus, there is the potential for the provision of a highlighting tool to foster intertextual integration but also a need to examine the conditions under which the use of such a tool may be most facilitative. Here, we examine how both highlights *and* annotations can support students' learning from multiple texts, for instances by supporting students' abilities to identify and track key concepts and integrate these across texts.

1.5. Promoting strategy use during multiple text learning

Analyses of both think-alouds and students notes have determined an initial pool of strategies that students use in learning from multiple texts and identified ones that may be particularly facilitative (List, 2020). Principally, these include students engaging in cross-textual linking or integration (Barzilai et al., 2021; McCrudden et al., 2022). Yet much of this prior work has been non-experimental in nature – that is the particular strategies that students have elected to use have been naturalistically captured, sans efforts to inculcate these or to encourage students to adopt particular modes of processing (Gerjets et al., 2011). Identifying means of fostering strategy use is essential for improving students' performance when learning from multiple texts.

Two means of encouraging particular forms of processing have been introduced in prior work. First, somewhat uniquely, Stadler and Bromme (2007) designed *met.a.ware*, as a note-taking system, prompting students to engage in source evaluation and comprehension monitoring when learning from multiple texts. Source evaluation was fostered by asking students to rate author expertise, bias, and their confidence in the information introduced, in association with each text that students took notes on. Metacognition was elicited by asking students to rate the quality of their knowledge and comprehension as a result of reading, and to rate how much information they still needed, in association with each text. Stadler and Bromme (2007) found students in the monitoring group to outperform a control group on a factual knowledge test, with no additional differences identified between groups and no cross-group

differences identified for subject matter comprehension nor for source recall. Elsewhere, systems designed to foster students' note-taking based on multiple texts have primarily focused on aiding students in attending to source (Britt & Aglinskas, 2002) or formulating source-content links (Barzilai et al., 2020).

Beyond designing note-taking interfaces to support multiple text learning, task assignment has also been used as a secondary means of fostering strategy use. In the literature on learning from multiple texts, assigning students to complete various types of task assignments (e.g., to write an argument vis-à-vis a summary) has repeatedly been found to result in differences in processing and performance (List et al., 2019; Schoor et al., 2021; Wiley & Voss, 1999). This is because tasks, in part, dictate what information students attend to during processing, what strategies students deploy, and what goals and standards students set for what constitutes adequate processing (McCrudden & Schraw, 2007). While traditionally, different types of task assignments have specified a desired product for students to achieve (e.g., to compose an argument or a research report), more recent work has assigned students to engage in different modes of processing when learning from multiple texts (List & Alexander, 2020; Bohn-Gettler & McCrudden, 2022).

Only one study, to our knowledge, has integrated these two approaches to fostering strategy use. Kobayashi (2009) examined students' external strategy use in response to two different task conditions, with students assigned either to form an opinion or to identify relations across six conflicting texts. When students were able to generate external notes (i.e., not assigned to a control group), they were found to highlight information, as well as to compose summaries, generate personal ideas (i.e., including elaborative and evaluative reactions), and to create intertextual markings. Kobayashi (2009) found students assigned to the relation formation task condition and allowed to generate external notes to perform better than students not allowed to produce external notes. However, no differences across external notes conditions were found among students assigned to form an opinion. Still, Kobayashi (2009) examined students' textual annotations rendered in print, not digitally, and the specific associations between the types of external notes that students generated and task performance was not examined.

1.6. Present study

Here, we build on Kobayashi's (2009) work by likewise investigating these two means of fostering students' strategy use during multiple text learning. First, we provide students with different task instructions directing them to engage in one of four different modes of strategic processing. Second, we support such strategic processing with the provision of a highlighting and annotation tool, to determine potential differences in processing and performance.¹ Although a constellation of strategies, including self-explanation, cross-textual linking, metacognition, and evaluation are likely needed for students to learn from multiple texts (List, 2020), here we aim to analytically induce these, separately, as distinct forms of processing. We do this for three primary reasons. First, we want to comparatively determine the advantages of particular strategic approaches, and document, within students' annotations, how particular strategic approaches manifest. Second, we want to consider whether some modes of processing may be easier to elicit through task instructions over others. Third, to the extent that some deep-level processing strategies (e.g., metacognition) manifest only to a limited extent when students engage these spontaneously, we seek to determine whether specifically cuing students to use these strategies can elicit them in greater volume and whether this elicitation results in differences in performance (Gerjets et al., 2011).

The four strategic approaches examined ask students to highlight and comment on any (a) *important* information in text (i.e., relevance processing condition); (b) *related* or *connected* information across texts (i.e., intertextual processing condition); (c) information that helped them judge or evaluate text *trustworthiness* (i.e., evaluation condition); or (d) information that was *confusing* or *difficult* for them to understand (i.e., metacognitive monitoring condition). The relevance condition served as the target comparison group for the other three conditions. This was done as relevance determinations have been identified as the foundational mechanism supporting multiple text processing in the MD-TRACE (Rouet & Britt, 2011). Still, such comparisons should be treated with caution as this study did not include a true control group. The intertextual processing condition was included given the preponderance of think-aloud and notes-analysis work finding integrative processing or cross-textual linking to be a key mechanism supporting learning from multiple texts (List et al., 2019; McCrudden et al., 2022).

Evaluation and metacognitive monitoring conditions were included because these have also been associated with positive multiple text learning outcomes, albeit to a more limited extent than cross-textual linking, and because these approaches to strategic processing were specifically used by Stadtler and Bromme (2007). Although metacognitive monitoring includes a variety of processes (Afflerbach & Cho, 2009), we specifically operationalized this as students' identification of problems in comprehension, as this has been the aspect of metacognition most commonly examined in studies of learning from multiple texts. Specifically, students were instructed to: "highlight and comment on any information that is confusing or difficult for you to understand." This is consistent with Afflerbach & Cho (2009) framework of constructively responsive reading defining metacognitive aspects of students' learning from multiple texts as including their, "detecting a comprehension problem" and "monitoring [the] degree and nature of comprehension" (pp. 80–81). Elsewhere, in capturing metacognition, Wolfe and Goldman (2005) asked students to report comprehension successes or failures. Linderholm et al. (2014) coded metacognition as students' reporting a lack of understanding, as did McCrudden et al. (2022). Thus, engaging students in metacognition by asking them to identify that information in texts that was confusing or difficult for them to

¹ We define processing as students' cognitive and behavioral engagement with texts throughout reading. We define performance as students' completion of an outcome measure, after reading, with this outcome measure tapping students' documents model construction or cognitive representation of all texts read (List et al., 2019). Although we separate processing and performance in the design of this study, we recognize that these are related and, as per the MD-TRACE, students may re-process texts when encountering difficulties in performance (Rouet & Britt, 2011).

understand seems largely consistent with how metacognition has been operationalized in prior work. Nevertheless, it is important to note that in providing students with such instructions, we may have increased metacognitive monitoring sans their being an expressed record of this. That is, students may not have found any of the information in texts challenging or difficult to understand and thus may not have highlighted or annotated the texts provided, despite engaging in metacognitive monitoring. This makes the metacognitive monitoring condition, in this study, distinct from the other three conditions examined.

To capture evaluation, in addition to examining students' highlights and annotations, hyperlinked source information (e.g., [1]) was embedded within each text. Consistent with prior work, students' accessing of embedded source information, during reading, was used as an additional measure of evaluation (List et al., 2017; Schoor et al., 2020).

We had the following research questions.

1.7. Research questions

1. Across conditions, what types of highlights and annotations do students produce? To what extent do they access references?

This was an exploratory research question to determine the types of highlights and annotations that students rendered. Based on prior work, we expected students to highlight and annotate key terms and other important information (Kalir, 2020) as well as to generate *linking* annotations, connection texts' content to prior knowledge and to other texts (Chen & Chen, 2014). Chen et al. (2020) also found students to generate metacognitive annotations, including indications of a lack of understanding. Still, to our knowledge, the content of highlights and annotations has only been coded within one multiple text study (Kobayashi, 2009) and can be expected to be somewhat idiosyncratic to the particular information provided in texts. Thus we approached determining categories for students' highlights and annotations in a largely bottom-up fashion.

2. Do students' highlights, annotations, and accessing of references differ across strategy conditions (i.e., assignment to relevance, intertextual linking, evaluative, or metacognitive monitoring conditions)?

We expected students assigned to the intertextual processing condition to generate more connections across texts during reading, followed by students assigned to the metacognitive monitoring and evaluation conditions, with students in the relevance processing condition generating the fewest cross-textual connections, overall.

3. Does students' multiple text comprehension and integration performance differ across strategy conditions (i.e., assignment to relevance, intertextual linking, evaluative, or metacognitive monitoring conditions)?

We expected students assigned to the intertextual processing condition to outperform all three other conditions with regard to the volume of integration included in writing; we expected students assigned to the evaluation and metacognition monitoring conditions to outperform students assigned to engage in relevance processing. Based on results from Stadtler and Bromme (2007), we expected students assigned to the metacognitive monitoring condition to outperform students assigned to the evaluation condition. Improved performance was expected to reflect students including more information and more integrated or connected information in writing and students demonstrating higher objective comprehension and integration performance. Still, this hypothesis is tentative as Stadtler and Bromme (2007) asked students to rate their comprehension for every text accessed, in the conditions including metacognitive monitoring, while students in this study were only asked to highlight and annotate texts if they found these confusing or difficult to understand.

4. Controlling for strategy condition, to what extent do students' highlights, annotations, and accessing of references predict multiple text comprehension and integration performance?

Consistent with prior work examining students' text annotations and think-alouds, we expected students demonstrating more cross-textual linking, as well as more evaluation and metacognitive monitoring, within their highlights and annotations, to evidence more integrative writing and better objective comprehension and integration performance. Consistent with other studies on learning from multiple texts (Barzilai et al., 2018; Hagen et al., 2014), multiple performance measures were included in this study to comprehensively and multidimensionally assess students' multiple text learning. In particular, we included both a writing measure and objective (i.e., selected response) measures, as different means of tapping students' learning. Through these measures, we assessed both single-text comprehension (or the representation of information from individual texts in students' writing) and multiple text integration. Integration, or students' connection formation across texts, has been found to be particularly essential for learning from multiple texts (List et al., 2021; List & Alexander, 2019).

2. Methods

2.1. Participants

Participants were 278 undergraduate students enrolled in a large university in the Northeastern United States (age: $M = 19.10$, $SD = 1.33$). This was a convenience sample. The sample was 72.94% ($n = 200$) female, 16.55% ($n = 46$) male, with 0.72% ($n = 2$) of

students identifying as non-binary. In the sample, 74.10% ($n = 206$) of students identified as White, 6.83% ($n = 19$) identified as biracial or mixed-race, 4.32% ($n = 12$) identified as Asian, 2.52% ($n = 7$) identified as Hispanic/Latino/a, and 0.72% of participants ($n = 2$) as African American/Black. Participants were primarily first- (48.20%, $n = 134$) and second-year (31.29%, $n = 87$) year students, pursuing education-related majors. Participants were recruited from an introductory educational psychology course and received research credit for participation. All students consented to participation. Some participants (10.79%, $n = 30$) did not report full or partial demographic information; percentages are calculated based on the total sample.

An a-priori power-analyses using G*Power (Faul et al., 2007) determined that a sample of 180 individuals would be needed to identify medium effects ($f = 0.25$) using an ANOVA, with four groups, when $\alpha = 0.05$, and $\beta = 0.80$, by convention.

2.2. Design

This study adopted a between-subjects design. Students were assigned to complete a multiple text reading-writing task under four different strategy conditions. These instructed them to engage in relevance-focused processing, intertextual linking, evaluation, or metacognitive monitoring.

2.3. Procedures

This study had three primary parts. First, students were asked to complete an assessment of prior knowledge. Second, students were asked to complete a multiple text reading-writing task. This involved their (a) receiving a set of task instructions in accordance with their randomly assigned strategy condition, (b) reading four texts addressing mass incarceration in the United States, with tools to support highlighting and annotating provided, and (c) writing a research report about the information read. Finally, students completed measures of multiple text comprehension and integration and reported demographic information. Students did not have access to the study texts, nor to their highlights and annotations, when composing their research reports nor when completing measures of comprehension and integration. Thus, these assessments functioned, in part, as recall tasks, capturing the nature and extent of students' cognitive representations of multiple texts, after reading. The study took participants approximately 1 hour to complete.

2.4. Materials

2.4.1. Task and experimental condition

All students received general task instructions: *In this study, we are going to ask you to read four texts to write a research report on mass incarceration in the United States*; as well as instructions specific to their assigned processing condition. These specific instructions directed students to highlight and comment on “any important information in texts” (i.e., relevance condition); “any information that is related or connected across texts” (i.e., intertext linking condition); “any information that helps you evaluate text trustworthiness” (i.e., evaluation condition); or “any information that is confusing or difficult for you to understand” (i.e., metacognitive monitoring condition). After receiving these specific task instructions, students also viewed screenshots explaining how to use the highlighting and annotating functions and how to access references within each text (i.e., by clicking embedded hyperlinks).

2.4.2. Texts

Four texts, addressing the topic of mass incarceration in the United States, were used in this study. Mass incarceration was selected as the topic of this task as it was a complex and controversial issue, about which various complementary and conflicting perspectives could be introduced. We further expected this topic to be somewhat familiar and of interest to students, without their having extensive conceptual knowledge about this, given their majors (i.e., in education).

Each text included three main arguments about mass incarceration, with a total of six distinct subtopics presented across the four texts introduced. This is because while arguments addressing two subtopics appeared uniquely within two of the texts, arguments

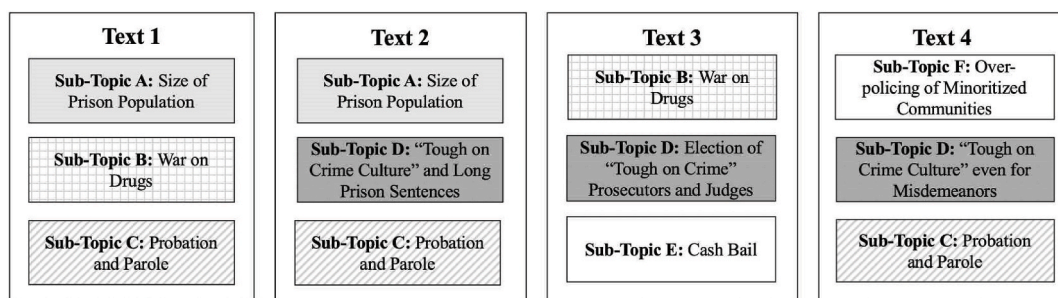


Fig. 1. Diagrammatic representation of arguments and subtopics across texts.

Note: Subtopics A and D presented complementary information across texts; Subtopics B and C presented conflicting information across texts; Subtopics E and F appeared uniquely in texts.

addressing the remaining four subtopics were distributed across either two or three texts, with either complementary or conflicting claims and evidence put forth, in each case. For example, two arguments addressing the subtopic of the role of drug-related arrests in mass incarceration discordantly suggested either that the War on Drugs and mandatory minimums played a large role in increasing mass incarceration or argued that those convicted of non-violent drug-related crimes represented only a minority of the total prison population. Thus, arguments on the same subtopic, like these, were purposefully constructed to appear across multiple texts in either a complementary or conflicting fashion because we were specifically interested in whether each of the four target strategy conditions examined would promote cross-textual linking. See Fig. 1 for a diagrammatic representation of arguments on different subtopics introduced across texts.

Each text was written in a neutral, academic tone and was between 253 and 258 words in length. Flesch-Kincaid grade-level ranged from 11.9 to 13.1, indicating that texts were appropriate for an undergraduate audience. Source information was provided, with each text described as a feature article in a popular-press publication and attributed to a professor affiliated with a prominent university in the United States. Each text also included three embedded references providing source information regarding the origin of statistical evidence included in texts. Students were able to access these references by clicking hyperlinks (e.g., [1]) in texts. Excerpts from all study texts are included in Appendix A.

2.4.3. Log data

As students accessed texts, a variety of log data were collected. Passive data logged included the amount of time that students spent on text access and the number of hyperlinked references they accessed. Student behaviors recorded included the number of highlights they rendered in each text, which text segments they highlighted, the number of text segments they annotated, and the content of their annotations. Appendix B includes a screenshot of the highlighting and annotating interface used.

2.4.4. Response composition

After being presented with all four texts and highlighting and annotating these, students were asked to write a research report on the information provided. Specifically, they received the following task instructions: *“Please write a research report about mass incarceration in the United States. Please try to include all of the key concepts and ideas from the texts. As you write, please try to identify connections across the texts you read. If you’re able, please cite sources in composing your response.”*

2.5. Measures

2.5.1. Prior knowledge

Students’ prior knowledge about the topic of mass incarceration was assessed using an eight-item term-identification measure. In particular, students were asked to define each of eight key terms (e.g., over-policing; misdemeanor; recidivism), related to mass incarceration and appearing in the study texts, in an open-ended fashion, with responses scored as correct or incorrect. Two raters coded 43 student responses (15.47%). Inter-rater reliability for the prior knowledge measure was Cohen’s $\kappa = 0.81$. Prior knowledge scores ($M = 3.28$, $SD = 1.87$) were not found to differ across conditions, $F(3, 267) = 0.05$, $p = 0.99$; thus, are presented descriptively and excluded from further analyses.

2.5.2. Highlighting

The information that students highlighted in text was coded in two ways. First, the total number of highlights that students rendered was recorded. These reflected 2283 distinct segments of text highlighted, on average, 14.99 words in length. Second, the type of information that students highlighted was categorized. Students highlighted five different types of information. This included (a) main ideas in texts, (b) supporting details, typically elaborating or providing examples of claims stated in main ideas, (c) definitions of key terms (e.g., recidivism, parole, probation), (d) statistical information provided in support of claims, and (e) source information, capturing information about texts’ origin (e.g., authors’ names, affiliations). Highlighted information that was either too brief to categorize (e.g., “rtionate”) or was otherwise not reflective of these categories was categorized as “other.” Students’ highlights were automatically categorized (i.e., via pattern matching) based on the segments of text selected. Highlighted clauses were placed into their correspondent categories regardless of whether these were only partially highlighted or selected in full. For example, students highlighting the clause, “in 2018 there were over 2.3 million people in U.S. prisons,” and the clause, “over 2.3 million people,” were both categorized as their highlighting statistical information. Thus, students received scores for both the total number of highlights

Table 1
Sample information highlighted in texts.

Highlight Category	Example of Text Highlighted
Main Ideas in Text	More people are incarcerated in the U.S. than anywhere else in the world (<i>Text 2</i>)
Supporting Details	These include paying supervision fees, meeting with a parole officer, submitting to drug testing, and attending specific programs (e.g., anger management) (<i>Text 4</i>)
Definitions of Key Terms	Probation, is a period of community supervision ordered by courts as a punishment, in lieu of jail time, for a criminal offense (<i>Text 1</i>)
Statistical Information	The average cost per parolee is \$1547 a year or \$4.24 per day, whereas the average annual per capita cost for imprisonment is \$14,603 or \$40 a day (<i>Text 1</i>)
Source Information	Dr. Mark Miller, Professor of Public Policy (<i>Text 3</i>)

they produced and for the number of main ideas, supporting details, key terms, statistical, and source information that they highlighted, across the four study texts. See Table 1 for samples of the text segments highlighted.

2.5.3. Annotations

When students highlighted specific information in texts, they were further provided with the option to add a comment or annotation to the selected information. A total of 785 annotations were coded and placed into one of six possible categories. These included (a) *paraphrases*, reflecting summaries or close restatements of text-based information; (b) *elaborations*, including self-explanations of information in text, based on prior knowledge, or students' connecting of information introduced in disparate sections of the same text to aid comprehension; (c) *categorizations*, or labeling of information in text as definitions, statistical evidence, or proposed causes or solutions; (d) *intertextual relations*, linking information in one text to previously read texts or identifying a common theme across texts; and (e) *evaluations*, reflecting judgements of either source trustworthiness or of the quality of information included in texts (e.g., its factual basis). All other annotations not belonging to these categories were coded as *other*; these included affective reactions and metacognitive judgements that were each reflected in less than five percent of annotations. All annotations were coded by both study authors, without condition information available. Inter-rater agreement was Cohen's $\kappa = 0.69$, with disagreements resolved through discussion. See Table 2 for sample annotations.

Annotation categories were developed using a four-step process. First, both authors reviewed the literature on common categories used to score students' annotations. Second, they independently reviewed and categorized a portion of students' annotations. Third, the categories that reviewers independently generated were corroborated, with categories added, removed, and clarified and with prototypical and boundary responses identified in association with each category. This independent review and corroboration process was repeated two more times, until a coding scheme was finalized. This was, then, applied to all students' annotations, with raters coding annotations independently and then corroborating their classifications.

2.5.4. References accessed

Log data were used to determine how many references students accessed, across texts.

2.5.5. Research report scoring

Students' research reports were scored in two ways. First, students received a score for content coverage, reflecting the number of arguments, from the 12 presented across texts, that they included in writing. The subtopic of these arguments was further noted, allowing us to capture whether students included complementary and conflicting arguments on the same subtopic, from across texts, in the responses they composed. Second, students received scores for cross-textual integration, reflecting the linking of distinct arguments or ideas across texts. While some of these inter-textual links were those intended by the design of study texts (e.g., students recognizing that texts presented conflicting perspectives on the role of drug-related arrests in mass incarceration); others were spontaneously formed by students in the study. Two raters (i.e., the first and second author) coded 124 (44.60%) of students' responses, without looking at participants' assigned condition. Cohen's kappa inter-rater agreement ranged from $\kappa = 0.56$ to $\kappa = 0.91$ for the identification of various subtopics in students' responses (i.e., students writing about the size of the prison population vis-à-vis cash bail); disagreements were resolved through discussion and by comparing the content in students' responses to original study texts to track information origin. A response excerpt is included in Appendix C.

2.5.6. Objective comprehension performance

In addition to composing research reports, students were asked to respond to 12 term identification items assessing comprehension. Each item provided students with a definition and asked them to supply the key term it referred to, with terms drawn from the four study texts (e.g., *minor offenses like turnstile jumping, shoplifting and trespassing, also referred to as 'quality of life crimes,' are called:*

Table 2
Sample annotations in texts.

Annotation Category	Text Segment	Annotations
Paraphrase	parole has become a major contributor to recidivism, or re-incarceration (Text 2)	many people in parole go back to prison
Elaboration	over-policing, or intense surveillance of minoritized communities by police. (Text 4)	this a problem that not everybody sees
Categorization	Probation, is a period of community supervision ordered by courts as a punishment, in lieu of jail time, for a criminal offense. (Text 4)	definition
Intertextual Relation	According to government statistics in 2018 there were over 2.3 million people in U.S. prisons. This corresponds to 698 people being incarcerated out of every 100,000 citizens, reflecting 1 out of every 143 U.S. adults (Text 1)	corresponds to the last 3 articles; the amount of people in prisons (ALOT)
Evaluation	Dr. Mark Miller, Professor of Public Policy (Text 3) Although seemingly a compassionate policy (Text 2)	Professor article. Usually trustworthy. Pointing out the opposing view or the rationale for decisions even when they disagree with them helps to solidify their own argument
Other	recidivism, or re-incarceration (Text 3)	word i didnt know

misdeemeanors). Thus, this assessment was intended to capture students' conceptual understanding emerging from the texts they read; such conceptual understanding may be understood as reflected in students' acquisition of disciplinary terms (Bolger et al., 2008; Joshi, 2005). More generally, vocabulary acquisition and comprehension have been conceptually and empirically linked (Baumann, 2009; Georgiou & Das, 2018), thus a vocabulary-focused measure was used to assess comprehension in this study. Responses were scored as correct or incorrect and the 12-item measure had a Cronbach's alpha reliability of 0.75.

2.5.7. Objective integration performance

Moreover, students were asked to complete a multiple text integration task. This provided students with 14 statements, reflecting correct or incorrect relations drawn across texts. Students were asked to determine whether each statement was correct or incorrect and, if marking a statement as incorrect, to provide an appropriate justification (although only students' item determinations, not justifications, were used in scoring). An *I don't know* option was also provided. Reliability for the 14-item measure was Cronbach's $\alpha = 0.74$.

3. Results

3.1. Research question 1: across conditions, what types of highlights and annotations do students produce? To what extent do they access references?

Research Question 1 was descriptive, examining the total number and types of highlights and annotations that students rendered. Students made, on average, 8.42 ($SD = 10.33$) distinct highlights across the four texts they read. These most commonly involved highlighting data or statistical information ($M = 3.34$, $SD = 4.43$) and key terms ($M = 2.47$, $SD = 3.67$) within each text. Students made, on average 2.90 annotations, across texts and conditions, indicating that students did not necessarily annotate each of the four documents they accessed. However, this average should be interpreted with caution as only approximately one-third of the sample (33.69%, $n = 94$) formulated any annotations during reading. Among those including at least one annotation, the average number of annotations formulated was 8.35 ($SD = 6.59$), across the four texts. Annotations most commonly focused on categorizing information ($M = 0.81$, $SD = 2.25$, e.g., identifying information as a definition, a statistic, or a main idea), as well as on paraphrasing ($M = 0.62$, $SD = 1.84$) and elaborating ($M = 0.71$, $SD = 1.95$) information in texts. Students' reference access scores ranged from zero to 12 ($M = 0.83$, $SD = 2.79$).

3.2. Data screening

Prior to inferential analyses, variables were screened for normality, with a number of extreme values for skewness and kurtosis identified. When skewness was greater than 2 and kurtosis was greater than 7 (Hair et al., 2010), variables were transformed by taking their square-root, with all transformed variables used in analyses, but with untransformed means and standard deviations reported in tables and in-text to aid interpretability. Skewness and kurtosis values for original and transformed variables are in Table 3. Variable transformation solved most, but not all, issues of non-normality. However, we proceeded with analyses as the F-test has been found to be robust to violations of normality (Box & Watson, 1962; Donaldson, 1968).

Correlations among the types of highlights and annotations that students produced are provided in Tables 4 and 5.

Table 3
Skewness and kurtosis values for all variables of interest.

Variable	Original Skewness	Original Kurtosis	Transformed Skewness	Transformed Kurtosis
Highlights				
Main Ideas	2.05 (0.15)	5.49 (0.30)		
Supporting Details	2.48 (0.15)	7.06 (0.30)	1.54 (0.15)	0.88 (0.30)
Key Terms	1.94 (0.15)	4.37 (0.30)		
Statistics	1.52 (0.15)	2.48 (0.30)		
Source Information	6.47 (0.15)	45.34 (0.30)	5.27 (0.15)	27.68 (0.30)
Total Highlights	1.58 (0.15)	3.29 (0.30)		
Annotations				
Paraphrases	3.99 (0.15)	17.87 (0.30)	2.46 (0.15)	5.39 (0.30)
Elaborations	3.36 (0.15)	11.47 (0.30)	2.25 (0.15)	4.08 (0.30)
Categorizations	3.52 (0.15)	13.52 (0.30)	2.28 (0.15)	4.33 (0.30)
Intertextual Relations	6.42 (0.15)	46.42 (0.30)	4.58 (0.15)	21.57 (0.30)
Evaluations	6.12 (0.15)	40.99 (0.30)	4.72 (0.15)	22.39 (0.30)
Total Annotations	2.19 (0.15)	4.55 (0.30)		
Total References Accessed	5.05 (0.15)	27.29 (0.30)	2.81 (0.15)	8.68 (0.30)
Performance Measures				
Total Arguments	0.26 (0.16)	0.14 (0.32)		
Integration in Writing	1.70 (0.16)	3.74 (0.32)		
Objective Comprehension	-0.31 (0.15)	-0.40 (0.31)		
Objective Integration	-0.50 (0.15)	0.36 (0.31)		

Note: Variables in gray were subjected to a square-root transformation to account for non-normality.

Table 4
Correlation among highlights.

	1.	2.	3.	4.	5.
1. Main Ideas	1.00				
2. Supporting Details ^a	0.53***	1.00			
3. Key Terms	0.72***	0.60***	1.00		
4. Statistical Info.	0.67***	0.58***	0.61***	1.00	
5. Source Info. ^a	-0.05	-0.10	-0.07	0.03	1.00

Note.

^a Indicates variables transformed via a square-root transformation to increase normality.

Table 5
Correlations among annotations.

	1.	2.	3.	4.	5.
1. Paraphrases ^a	1.00				
2. Elaborations ^a	0.66***	1.00			
3. Categorizations ^a	0.45***	0.50***	1.00		
4. Intertextual Relations ^a	0.26***	0.34***	0.21***	1.00	
5. Evaluations ^a	-0.04	0.10	0.001	0.06	1.00

Note.

^a Indicates variables transformed via a square-root transformation to increase normality.

3.3. Research question 2. do students' highlights, annotations, and accessing of references differ across strategy conditions?

The second research question examined whether students' multiple text processing differed by strategy condition. Three indicators of processing (i.e., students' highlights, annotations, and references accessed) were examined.

3.3.1. Highlights

The total number of highlights that students rendered was found to differ significantly by strategy condition, $F(3, 267) = 13.14$, $p < 0.001$, $\eta^2 = 0.13$. Post-hoc, pair-wise comparisons using Tukey's HSD determined that students assigned to the metacognitive monitoring condition ($M = 2.09$, $SD = 4.08$) highlighted significantly less information than students assigned to the other three conditions [relevance processing: $M = 9.79$, $SD = 10.90$; intertextual linking: $M = 10.84$, $SD = 10.99$; evaluation: $M = 11.01$, $SD = 10.95$] ($ps < 0.001$).

We further examined whether the number of highlights, of different types, that students rendered differed by strategy condition. This determined that in addition to having fewer highlights, overall, students assigned to the metacognitive monitoring condition highlighted significantly fewer main ideas [$F(3, 267) = 10.38$, $p < 0.001$, $\eta^2 = 0.10$], definitions [$F(3, 267) = 7.15$, $p < 0.001$, $\eta^2 = 0.07$], and instances of statistical information [$F(3, 267) = 12.83$, $p < 0.001$, $\eta^2 = 0.13$], as compared to students in all three of the other strategy conditions. Additionally, students in the metacognitive monitoring condition highlighted significantly fewer supporting details than students in the relevance processing and intertextual linking task conditions, [$F(3, 267) = 3.53$, $p < 0.05$, $\eta^2 = 0.04$]. Finally, students in the evaluation strategy condition highlighted significant more source information than students in the other three conditions, [$F(3, 267) = 11.91$, $p < 0.001$, $\eta^2 = 0.12$]. Means and standard deviations are presented in Table 6.

Table 6
Descriptives of highlights across texts.

Information Highlighted	Relevance Processing	Inter-Textual Linking	Evaluation	Metacognitive Monitoring	Total
	M (SD)	M (SD)	M (SD)	M (SD)	M (SD)
Main Ideas	2.04 (2.33) ^c	2.49 (2.93) ^a	2.10 (3.06) ^b	0.34 (0.80) ^{abc}	1.74 (2.57)
Supporting Details ^a	0.41 (0.74) ^a	0.40 (0.78) ^b	0.30 (0.55)	0.12 (0.41) ^{ab}	0.31 (0.64)
Key Terms	3.04 (3.65) ^b	3.43 (4.23) ^a	2.58 (3.94) ^c	0.82 (1.89) ^{abc}	2.47 (3.67)
Statistical Info.	4.03 (4.66) ^c	4.21 (4.62) ^b	4.51 (4.83) ^a	0.65 (1.74) ^{abc}	3.34 (4.43)
Source Information ^a	0.00 (0.00) ^a	0.00 (0.00) ^b	1.09 (2.89) ^{abc}	0.00 (0.00) ^c	0.27 (1.50)
Total Highlights	9.79 (10.90) ^c	10.84 (10.99) ^b	11.01 (10.95) ^a	2.09 (4.08) ^{abc}	8.42 (10.33)

Note: Significant differences are noted with superscripts. Superscripts should be read horizontally, within each row. For example, Row 1 can be interpreted as students in the metacognitive monitoring condition highlighting significantly fewer main ideas than students in the ^ainter-textual linking, ^bevaluation, and ^crelevance processing conditions. Superscripts are ordered based on the magnitude of each mean difference.

^a Indicates variables transformed via a square-root transformation to increase normality in ANOVA analyses; however, means are reported in untransformed form to aid interpretability.

3.3.2. Annotations

We further examined whether the total number of annotations rendered differed by strategy condition. Significant differences were found, $F(3, 267) = 2.97, p < 0.05, \eta^2 = 0.03$. Post-hoc, pair-wise comparisons using Tukey's HSD again found students assigned to the metacognitive monitoring condition to produce significantly fewer annotations ($M = 1.26, SD = 2.91$) than students in the evaluation condition ($M = 3.96, SD = 6.21$), $p < 0.05$.

We further examined whether there were differences in the types of annotations that students produced across strategy conditions. Indeed, significant differences were found in the number of paraphrased [$F(3, 267) = 2.68, p < 0.05, \eta^2 = 0.03$], evaluative [$F(3, 267) = 16.28, p < 0.001, \eta^2 = 0.16$] and intertextual [$F(3, 267) = 2.63, p = 0.050, \eta^2 = 0.03$] annotations that students produced. Planned, post-hoc pairwise comparisons using Tukey's HSD found students in the intertextual linking strategy condition to produce significantly more restatements ($M = 0.97, SD = 2.24$) and intertextual annotations ($M = 0.37, SD = 1.26$) than students in the metacognitive monitoring task condition (restatements: $M = 0.22, SD = 1.08$; intertextual links: $M = 0.01, SD = 0.12$), $ps < 0.05$. Moreover, students assigned to the evaluation strategy condition produced significantly more evaluative annotations ($M = 1.57, SD = 3.74$) than students in the other three task conditions ($Ms = 0.00, SDs = 0.00$), $ps < 0.001$. No other significant differences in the types of annotations that students generated were identified, $ps > 0.07$. See Table 7 for descriptive information.

3.3.3. References accessed

Finally, as a third indicator of processing, we examined whether the number of references that students accessed during reading differed across the four strategy conditions. Significant differences were found, $F(3, 267) = 9.22, p < 0.001, \eta^2 = 0.09$. In particular, follow-up analyses using Tukey's HSD determined that students assigned to the evaluation condition accessed significantly more references ($M = 2.30, SD = 5.00$) than students in the relevance processing ($M = 0.32, SD = 0.72$), intertextual linking ($M = 0.25, SD = 0.70$), and metacognitive monitoring ($M = 0.46, SD = 1.66$) strategy conditions, $p < 0.001$. Descriptive information is in Table 7.

3.4. Research question 3: Does students' multiple text performance differ by strategy condition?

Multiple text task performance was assessed by examining the number of arguments reflected in students' written responses and the extent to which these were integrated, as well as by examining objective measures of multiple text comprehension and integration.

3.4.1. Research report comprehensiveness

First, a one-way ANOVA was used to examine whether the number of arguments included in students' written responses, from the 12 introduced across texts, differed by strategy condition. However, no significant differences were found, $F(3, 232) = 0.82, p = 0.48$.

3.4.2. Research report integration

Likewise, the volume of integration or cross-textual linking reflected in students' responses did not significantly differ across conditions, $F(3, 232) = 1.47, p = 0.22$.

3.4.3. Objective comprehension performance

We further examined whether strategy condition was associated with performance on an objective measure of multiple text comprehension. However, no significant differences were identified, $F(3, 245) = 2.31, p = 0.08$.

3.4.4. Objective integration performance

Objective multiple text integration performance was also found not to differ by strategy condition, $F(3, 245) = 0.23, p = 0.88$.

Table 7

Descriptives of annotations and references accessed across texts.

Annotations	Relevance Processing	Inter-Textual Linking	Evaluation	Metacognitive Monitoring	Total
	M (SD)	M (SD)	M (SD)	M (SD)	M (SD)
Paraphrases ^a	0.50 (1.26)	0.97 (2.24) ^a	0.79 (2.36)	0.22 (1.08) ^a	0.62 (1.84)
Elaborations ^a	0.88 (2.22)	0.97 (2.42)	0.66 (1.77)	0.32 (1.07)	0.71 (1.95)
Categorizations ^a	1.44 (3.15)	0.66 (1.85)	0.63 (1.68)	0.50 (1.90)	0.81 (2.25)
Intertextual Relations	0.12 (0.59)	0.37 (1.26) ^a	0.07 (0.26)	0.01 (0.12) ^a	0.14 (0.72)
Evaluations	0.00 (0.00) ^a	0.00 (0.00) ^b	1.57 (3.74) ^{abc}	0.00 (0.00) ^c	0.39 (1.97)
Total Annotations	3.25 (5.94)	3.13 (6.20)	3.96 (6.21) ^a	1.26 (2.91) ^a	2.90 (5.55)
References Accessed	0.32 (0.72) ^b	0.25 (0.70) ^c	2.30 (5.00) ^{abc}	0.46 (1.66) ^a	0.83 (2.79)

Note: Significant differences are noted with superscripts. Superscripts should be read horizontally, within each row. For example, Row 1 can be interpreted as students in the metacognitive monitoring condition producing significantly fewer annotations including paraphrases than students in the ^ainter-textual linking condition. Superscripts are ordered based on the magnitude of each mean difference.

^a Indicates variables transformed via a square-root transformation to increase normality in ANOVA analyses; however, means are reported in untransformed form to aid interpretability.

3.5. Research question 4: controlling for strategy condition, to what extent do students' highlights, annotations, and accessing of references predict multiple text comprehension and integration performance?

For research question 4, multiple text processing indicators (i.e., highlights, annotations, references accessed) were used to predict each of the four multiple text outcomes of interest (i.e., the number of arguments included in students' research reports, research report integration, objective comprehension, and objective integration). Due to the number of processing variables captured through students' highlights, annotations, and references accessed, a backwards regression approach was used to statistically optimize predictor selection and model fit. Predictors entered in Step 1 included (a) strategy condition, with intertextual linking, evaluation, and metacognitive monitoring conditions indicator coded, relative to the relevance processing condition, (b) students' highlighting of main ideas, supporting details, definitions of key terms, and source and statistical information in texts, (c) students' paraphrased, elaborated, intertextual, and evaluative annotations, and annotations categorizing information in texts, and (d) the number of references that students accessed during multiple text use. Thus, a total of 14 predictors were entered in Step 1. No issues of multicollinearity were identified for any of the models presented (Tolerance > 0.73; VIF < 1.37).

3.5.1. Research report comprehensiveness

First, the model predicting the number of arguments included in students' research reports was significant both at the first, $F(14, 221) = 3.01, p < 0.001, R^2_{adj} = 0.11$, and last step, $F(5, 230) = 7.68, p < 0.001, R^2_{adj} = 0.12$, of the backwards regression. Individually significant predictors retained in the last step of the model were students' assignment to the evaluation condition ($\beta = -0.17, p < 0.05$), relative to the relevance processing condition, with this associated with lower research report comprehensiveness; as well as students highlighting key terms ($\beta = 0.19, p < 0.05$) and source information ($\beta = 0.16, p < 0.05$) in texts and accessing more references during reading ($\beta = 0.16, p < 0.05$). See Table 8 for a model summary.

3.5.2. Research report integration

Likewise, the backwards regression predicting research report integration was significant at both the first, $F(14, 221) = 1.80, p < 0.05, R^2_{adj} = 0.05$, and last step, $F(5, 230) = 4.50, p < 0.001, R^2_{adj} = 0.07$, of the model. Significant predictors retained in the final model included students' assignment to the inter-textual linking ($\beta = 0.18, p < 0.01$) and metacognitive monitoring ($\beta = 0.16, p < 0.05$) task conditions, with this assignment advantageous relative to students' membership in the relevance processing condition, as well as students highlighting of supporting details ($\beta = 0.14, p < 0.05$) and source information ($\beta = 0.15, p < 0.05$) in texts. See Table 9 for a model summary.

3.5.3. Objective comprehension performance

The backwards regression predicting objective comprehension performance was also significant in the first, $F(14, 234) = 3.20, p < 0.001, R^2_{adj} = 0.11$, and last, $F(4, 244) = 8.98, p < 0.001, R^2_{adj} = 0.11$, steps of the model. Individually significant predictors retained in the final step of the model included students' membership in the metacognitive monitoring condition ($\beta = 0.22, p < 0.001$), relative to the relevance processing condition, as well as students' highlighting of statistical information in texts ($\beta = 0.17, p < 0.05$), formulating

Table 8
Model predicting the number of arguments included in students' responses.

	B	SE(B)	β	Sig.
Initial Model				
<i>Strategy Condition</i>				
InterText Condition	0.41	0.40	0.08	0.30
Evaluation Condition	-0.62	0.43	-0.12	0.15
MetaCog Condition	0.66	0.41	0.13	0.11
<i>Highlights</i>				
Main Ideas	0.03	0.08	0.04	0.72
Supporting Details [^]	0.47	0.37	0.10	0.20
Key Terms	0.08	0.06	0.14	0.19
Statistical Information	0.04	0.05	0.08	0.38
Source Information [^]	0.68	0.40	0.17	0.09
<i>Annotations</i>				
Paraphrases [^]	0.16	0.27	0.05	0.55
Elaborations [^]	0.02	0.26	0.01	0.94
Categorizations [^]	-0.04	0.22	-0.02	0.85
Intertextual Relations [^]	0.06	0.38	0.01	0.87
Evaluations [^]	0.04	0.35	0.01	0.91
References Accessed [^]	0.36	0.20	0.14	0.07
Final Model				
Evaluation Condition	-0.88	0.36	-0.17	0.01*
Supporting Details [^]	0.58	0.34	0.13	0.09
Key Terms	0.11	0.05	0.19	0.01*
Source Information [^]	0.65	0.30	0.16	0.03*
References Accessed [^]	0.42	0.18	0.16	0.02*

Note: All beta coefficients should be interpreted with caution and not directly compared, as predictors indicated via [^] were transformed.

Table 9
Model predicting integration in students' written responses.

	<i>B</i>	<i>SE(B)</i>	β	Sig.
Initial Model				
<i>Strategy Condition</i>				
InterText Condition	0.55	0.32	0.14	0.09
Evaluation Condition	−0.24	0.35	−0.06	0.50
MetaCog Condition	0.50	0.33	0.13	0.13
<i>Highlights</i>				
Main Ideas	−0.03	0.07	−0.04	0.67
Supporting Details [^]	0.39	0.29	0.11	0.19
Key Terms	0.06	0.05	0.12	0.25
Statistical Information	−0.01	0.04	−0.03	0.76
Source Information [^]	0.57	0.32	0.18	0.07
<i>Annotations</i>				
Paraphrases [^]	−0.10	0.21	−0.04	0.66
Elaborations [^]	0.14	0.21	0.07	0.50
Categorizations [^]	−0.16	0.17	−0.08	0.35
Intertextual Relations [^]	0.20	0.31	0.05	0.51
Evaluations [^]	−0.03	0.28	−0.01	0.90
References Accessed [^]	0.25	0.16	0.13	0.11
Final Model				
InterText Condition	0.74	0.27	0.18	0.01**
MetaCog Condition	0.65	0.28	0.16	0.02*
Supporting Details [^]	0.48	0.23	0.14	0.04*
Sourcing Information [^]	0.49	0.23	0.15	0.04*
References Accessed [^]	0.27	0.14	0.14	0.06

Note: All beta coefficients should be interpreted with caution and not directly compared, as predictors indicated via [^] were transformed.

annotations categorizing information ($\beta = 0.17, p < 0.01$), and accessing references during reading ($\beta = 0.17, p < 0.01$). See Table 10 for a model summary.

3.5.4. Objective integration performance

The backward regression model predicting objective integration performance was significant at both the first, $F(14, 234) = 3.22, p < 0.001, R^2_{adj} = 0.11$, and last, $F(5, 243) = 7.93, p < 0.001, R^2_{adj} = 0.12$, steps. Individually significant predictors in the last step of the model included students' membership in the metacognitive monitoring condition ($\beta = 0.14, p < 0.05$), relative to the relevance processing strategy condition, as well as students' highlighting of statistical information ($\beta = 0.22, p < 0.01$), formulation of annotations categorizing information ($\beta = 0.14, p < 0.05$), and accessing of reference information ($\beta = 0.15, p < 0.05$) during processing. See Table 11 for a model summary.

Table 10
Models predicting objective comprehension performance.

	<i>B</i>	<i>SE(B)</i>	β	Sig.
Initial Model				
<i>Strategy Condition</i>				
InterText Condition	−0.79	0.49	−0.12	0.11
Evaluation Condition	−1.04	0.53	−0.16	0.050*
MetaCog Condition	0.97	0.51	0.15	0.06
<i>Highlights</i>				
Main Ideas	−0.13	0.11	−0.12	0.23
Supporting Details [^]	0.24	0.46	0.04	0.60
Key Terms	0.10	0.08	0.13	0.19
Statistical Information	0.11	0.06	0.17	0.07
Source Information [^]	0.10	0.50	0.02	0.84
<i>Annotations</i>				
Paraphrases [^]	0.15	0.32	0.04	0.65
Elaborations [^]	0.05	0.33	0.02	0.87
Categorizations [^]	0.33	0.27	0.10	0.22
Intertextual Relations [^]	−0.01	0.48	−0.001	0.99
Evaluations [^]	0.50	0.44	0.11	0.26
References Accessed [^]	0.44	0.25	0.13	0.08
Final Model				
MetaCog Condition	1.46	0.42	0.22	<0.001***
Statistical Information	0.11	0.04	0.17	0.01*
Categorizations [^]	0.58	0.21	0.17	0.01**
References Accessed [^]	0.56	0.20	0.17	0.01**

Note: All beta coefficients should be interpreted with caution and not directly compared, as predictors indicated via [^] were transformed.

Table 11
Models predicting objective integration performance.

	<i>B</i>	<i>SE(B)</i>	β	Sig.
Initial Model				
<i>Strategy Condition</i>				
InterText Condition	−0.45	0.52	−0.07	0.39
Evaluation Condition	−0.86	0.56	−0.12	0.13
MetaCog Condition	0.71	0.54	0.10	0.19
<i>Highlights</i>				
Main Ideas	0.07	0.11	0.06	0.54
Supporting Details [^]	0.15	0.48	0.03	0.76
Key Terms	−0.00	0.08	−0.001	1.00
Statistical Information	0.11	0.06	0.17	0.07
Source Information [^]	0.85	0.53	0.15	0.11
<i>Annotations</i>				
Paraphrases [^]	0.26	0.34	0.07	0.45
Elaborations [^]	−0.01	0.35	−0.002	0.98
Categorizations [^]	0.28	0.28	0.08	0.32
Intertextual Relations [^]	0.62	0.51	0.08	0.22
Evaluations [^]	0.10	0.47	0.02	0.83
References Accessed	0.51	0.26	0.15	0.05
Final Model				
MetaCog Condition	0.95	0.45	0.14	0.04*
Statistical Information	0.14	0.05	0.22	0.002**
Source Information [^]	0.67	0.39	0.12	0.09
Categorizations [^]	0.50	0.23	0.14	0.03*
References Accessed [^]	0.53	0.24	0.15	0.03*

Note: All beta coefficients should be interpreted with caution and not directly compared, as predictors indicated via [^] were transformed.

3.6. Exploratory models

We were principally interested in predicting multiple text comprehension and integration outcomes by examining which information students highlighted and annotated during reading. However, given recent work pointing to the importance of behavioral engagement when learning from multiple texts (List et al., 2021; Bråten et al., 2018) and the exploratory nature of this study, we also ran a complementary series of models using just quantitative indicators of highlighting and annotation to predict each performance outcome of interest. These analyses are included in Appendix D.

4. Discussion

Contributing to research seeking to improve students' learning from multiple texts, this study investigated whether multiple text learning could be facilitated by asking students to highlight and annotate four expository texts, addressing mass incarceration in the United States. This study further examined learners' highlights and annotations under four different strategy conditions, with students' digital annotations not previously directed in this way. This study makes contributions through both descriptive and inferential analyses. Each of these is discussed in turn.

4.1. Descriptives of students' highlights

Descriptively, this study is a robust examination of students' highlighting and textual annotations when learning from multiple texts. Although students highlight and annotate e-texts frequently through the course of everyday reading (Welsen et al., 2020), such highlights and annotations have been insufficiently examined in prior work on multiple text learning. This is despite the rich potential of these data to provide insights into students' processing. Prior studies have examined students' highlighting and annotations within the context of learning from single texts (Goodwin et al., 2020; Novak et al., 2012). However, this study is unique in analyzing highlights and annotations generated under various processing conditions and in association with multiple text comprehension and integration outcomes (see also Leroy et al., 2021; Leroy & Kammerer, 2022). Students were found to most commonly highlight (a) statistical information, serving as supporting evidence, (b) definitions of key terms, and (c) main ideas in texts. This finding seems aligned with prior work finding students, at least at the undergraduate level, to be fairly successful at rendering relevance determinations or identifying important information in texts (Anmarkrud et al., 2013). Students' frequent highlighting of statistical information, in particular, was an encouraging finding as this may indicate students' reasoning about text-based evidence, with such reasoning an essential yet underexamined competency (Du & List, 2022).

4.2. Descriptives of students' annotations

In terms of annotations, these were found to most frequently reflect students categorizing of information appearing in texts. This included students labeling information as putting forth a definition, introducing a statistic, or identifying a problem or proposing a

solution. Information categorization has also been identified as a common annotation type in prior work examining students' marking of single texts. For instance, [Simpson and Nist \(1990\)](#) trained students in annotation and encouraged them to develop a personal code to use when marking textbook passages from psychology, sociology, and history. Among the annotations that students generated were those identifying ideas (e.g., causes, effects), marking key words, and labeling examples – with all of these annotations potentially exemplifying categorization. [Porter-O'Donnell \(2004\)](#), in teaching high-school students how to annotate literary texts, taught students to distinctively mark characters, setting, and vocabulary in texts – suggesting the genre-specific nature of categorizing annotations. Within our study, introducing students to multiple expository texts, identifying key terms and recognizing supporting evidence for claims, in the form of statistics, seemed to be the processes most engaged during reading (at least as captured through learners' annotations). Identifying these specific elements in texts may have served to reduce the volume of information that students needed to make sense of or contend with during processing or may have reflected students' exploitation of the recording function of notes, allowing information to be revisited after reading ([Kiewra, 1989](#)) (although annotations were not available to students when they composed responses, in this study). Alternately, labeling information may have supported deeper-level processing, with students further elaborating that information they categorized, even if this elaboration was not reflected within the annotations that they produced.

4.3. Research question 2: highlights and annotations across strategy conditions

In addition to descriptively analyzing students' highlights and annotations, we were further interested in examining these across prompted strategy conditions. In terms of processing indicators, one persistent finding seemed to emerge – namely that placement into the metacognitive monitoring condition, asking learners to attend to, to highlight, and to annotate that information that was difficult for them to understand, was associated with fewer highlights and fewer annotations generated, across categories. On the surface these findings seem inconsistent with findings from [Stadler and Bromme \(2007\)](#) who found advantages for students' placement in the metacognitive monitoring condition, at least as far as factual knowledge was concerned. However, [Stadler and Bromme \(2007\)](#) did not examine indicators of processing. In this study, students' highlights and annotations may simply have been more limited in the metacognitive monitoring condition because students found all four of the texts introduced to be fairly accessible or easy to comprehend, obviating their need to highlight information that was confusing or difficult. This latter explanation is further borne out by there being no performance differences across strategy conditions, despite processing indicators being considerably more limited for students belonging to the metacognitive monitoring strategy condition. Moreover, echoing results from [Stadler and Bromme \(2007\)](#), membership in the metacognitive monitoring condition, relative to the relevance processing strategy condition, was predictive of objective comprehension and integration performance and students' integration in writing.

4.4. Research questions 3 & 4: examining performance across strategy conditions and predicting performance

Although no performance differences in neither comprehension nor integration emerged across strategy conditions, regardless of whether these were assessed objectively or through writing, performance on these outcomes was predicted by various indicators of processing. Patterns in *which* processing indicators were associated with performance are discussed. To start, students' highlighting of source information in texts was predictive of the number of arguments included in writing and the extent to which these were integrated, as well as of objective integration performance. This pattern of findings can be explained in at least two complementary ways. First, highlighting of source information, supporting students' abilities to judge or evaluate information quality, may have reflected deep-level processing, either supporting encoding on its own or indicating the use of additional, deep-level, comprehension-supporting strategies during reading. Second, from a Documents Model standpoint ([Britt et al., 1999](#); [Perfetti et al., 1999](#)), attending to source information, in particular, may have helped students “tag” or track arguments across texts, thereby supporting integration. Both of these explanations, considering the evaluative and representational functions of source information, are further supported by students' accessing of reference information in texts predicting argument inclusion in writing and objective comprehension and integration performance.

Beyond results broadly suggesting that what information students highlighted mattered for performance, learners' annotations were found to play a more limited role. In part, this may have been for reasons of design. That is, because students had to highlight information before they could record annotations, any variance explained by students' engagement in annotations may already have been captured within the information they highlighted. Additionally, the lack of a strong emergence of annotations-based indicators as predictors of performance may also have been due to the majority of the sample (63.67%, $n = 177$), not producing any annotations at all. Still, students' production of annotations categorizing information in texts was associated with objective comprehension and objective integration performance. This suggests that these annotation types helped students identify and encode key concepts and ideas in texts and link these across texts, resulting in comprehension and integration learning outcomes. These results are important as information categorization, despite constituting a fairly commonly produced annotation type in studies of single texts (e.g., [Simpson and Nist, 1990](#); [Zywica & Gomez, 2008](#)), has typically not been reported in think-aloud studies of multiple texts. Thus, this may be a strategy supporting multiple text learning that is difficult for students to enact without the provision of an annotation tool or a scaffold of some other kind ([Kobayashi, 2009](#)).

4.5. Impact and implications

Students' assignment to strategy condition did not, on its own, differentiate performance. Still, each of the four multiple text

learning outcomes examined was able to be predicted by students' highlighting of specific information in texts (i.e., definitions, statistics, source information), and in some instances, annotations and membership in intertextual linking and metacognitive monitoring strategy conditions, relative to the relevance processing group. These results offer at least three key insights for future work. First, highlights, and to a lesser extent annotations, emerged as both promising research tools for capturing processing and as reading scaffolds that may help students more effectively manage the volume of information inherent in learning from multiple texts. The affordances of highlights and annotations in supporting learning from multiple texts has previously been examined only to a limited extent (Kobayashi, 2009; Leroy & Kammerer, 2022). Indeed, as demonstrated by the range of highlights and annotations that students produced, there is a great variety of information, serving different functions (e.g., introducing main ideas, informing evaluation), that students need to contend with when processing even brief, academic texts. Highlighting and annotating, as demonstrated here, may be ways of helping students attend to and analyze these functions. This finding is important as highlighting and annotating texts have not always been found to improve learning (Goodwin et al., 2020; Inie et al., 2021); however, these strategies may be particularly facilitative for multiple text processing, given its increased demand for information management and conceptual linking (List & Alexander, 2018; List et al., 2021; Yukhymenko-Lescroart et al., 2022).

Second, highlights of statistical and source information, as well as annotations categorizing information in texts, were predictive of multiple text writing performance and objective measures of comprehension and integration. These types of highlights and annotations were emblematic of deep-level processing, or the connecting, transforming, and manipulation of information in texts (i.e., in contrast to recall-focused processing, Dinsmore & Alexander, 2016). Here, we demonstrate that students' generation of these types of highlights and annotations can be fostered, at least in part, through the task instructions introduced, encouraging students to engage in particular modes of processing. This constitutes an important finding as prior work examining digital annotations has largely not directed students to render annotations of particular types (Krouska et al., 2018). An important direction for future work is examining whether training students on how to better formulate highlights and annotations can increase the quality and number of these (Simpson & Nist, 1990), with deep-level highlights and annotations, in turn, perhaps through their manifest nature, improving performance. In other words, while strategy training has generally been found to improve reading comprehension outcomes (Graesser et al., 2005; McNamara, O'Reilly, Best, & Ozuru, 2006), using highlights and annotations as a mechanism for helping students implement and evidence the strategies taught may be a promising future direction for fostering multiple text learning.

Third, echoing findings from previous work (Stadtler & Bromme, 2007), assignment to the metacognitive monitoring condition, although associated with the more limited generation of highlights and annotations, was predictive of integration in writing and objective comprehension and integration performance, at least as compared to students assigned to engage in relevance-focused processing. This suggests the importance of integrating metacognitive monitoring into interventions seeking to improve students' learning from multiple texts (Greene et al., 2018). As suggested by the quality of students' written responses, such interventions – particularly those focused on improving students' integration of information across multiple texts – are very much in need (List et al., 2021; Barzilai et al., 2018).

4.6. Instructional implications

Although preliminary, the pragmatic implications of this study for instructors, seeking to improve students' learning from multiple texts, are two-fold. First, we find suggestive evidence that asking students to engage in highlighting and annotating may be an effective strategy for fostering multiple text learning, particularly if students are provided with explicit instruction in how to do so effectively. Second, consistent with prior work (Stadtler & Bromme, 2007), encouraging students to engage in metacognitive monitoring or in detecting comprehension problems during reading may be an effective means of fostering learning; and one that may be supported through the provision of a highlighting and annotation tool.

4.7. Limitations

Despite the strengths of this study, a number of limitations must be acknowledged. First, a key limitation comes from the non-representative nature of our sample. In particular, predominantly White females, majoring in education, and enrolled in their first or second year of university study, were examined. This requires that this work be replicated with additional student samples. In particular, the topic of this task – mass incarceration in the United States – was not one directly relevant to most participants' major. Therefore, replicating this study with a higher knowledge sample also stands as a direction for future work and would contribute to the generalizability of the study's findings. Beyond this, gender differences have sometimes been found in students' performance on multiple text tasks (see Anmarkrud et al., 2022 for a review; Barzilai et al., 2022). Additionally, we did not assess the extent of individuals' experience with the carceral and criminal justice systems in this study; investigating how students read and critically reason about issues of more and less personal and social relevance to them represents an essential direction for future work.

Additionally, a limitation stems from the few individual difference factors collected. Although we captured students' prior knowledge and found this not to differ across task conditions, additional measures, including epistemic beliefs, attitudes, and interest likely played a role in students' processing and were not examined (List et al., 2019; Bråten et al., 2011). Moreover, students' reading abilities were not assessed. Given that they were enrolled in a selective college, we expected the students in our sample to be competent readers. At the same time, many students have been found to struggle with college-level reading (Feller et al., 2020; Magliano et al., 2022) and reading comprehension has been associated with students' learning from multiple texts (Anmarkrud et al., 2022). Thus, there is a need to assess this competency in future work. Likewise, no validity check was introduced to ensure that our interpretations of students' highlights and annotations were accurate. Thus, a next step is to ask students to explain their annotations and highlights,

including why they formulated these and how they expected these to aid learning.

Likewise, students completed their highlights and annotations independently. Prior work has examined how highlights and annotations function when these are presented to students (i.e., rather than rendered by learners) and when these are collaboratively completed (Krouska et al., 2018; Novak et al., 2012; Samuel et al., 2011). So, there is a need to examine students' use of expert and social annotations, specifically when these are presented to support learning from multiple texts. Relatedly, the outcome measures used in this study were all completed without students having access to the texts provided nor to their highlights and annotations rendered. Although this was done to tap the degree of students' (cognitive) documents model construction after reading, this limited the ecological validity of the study. That is, in the real world, students often have access to their notes and reading materials when completing academic tasks. Accordingly, examining students' use of digital highlights and annotations for writing constitutes an important direction for future work, particularly given the hypothesized "storage," function of notes in facilitating learning (Kiewra, 1989).

Moreover, due to the volume of data collected for this study, a backwards regression approach was adopted to select predictors and optimize model fit. However, backwards regression carries with it a number of limitations, due to its exploratory nature (Derksen & Keselman, 1992; Thayer, 2002). Thus, future work, should, on the one hand, adopt more confirmatory analyses; and, on the other hand, consider using and cross-fitting additional prediction techniques (e.g., LASSO regression) for model specification. An additional methodological limitation comes from the fact that no control group, not provided with a highlighting and annotation tool, was included in this study. This decision was made on our part given that substantive work has already been conducted comparing the provision of highlighting and annotating options or not (Ben-Yehudah & Eshet-Alkalai, 2014, 2018; Kobayashi, 2009; Leroy et al., 2021). Rather than examining the absolute utility of such tools and given their ubiquity, we instead sought to examine how their use could be rendered more effective. Still, including a control condition certainly would have helped further contextualize some of the results from this study, as would adding a more neutral condition, wherein students were provided a highlighting and annotation tool, but were not given additional strategy instructions. In addition to including a control group, our operationalization of the various strategy conditions used in this study need additional conceptualizing in future work. In particular, students assigned to the metacognitive monitoring condition were only asked to identify comprehension difficulties. This likely resulted in their generating fewer highlights and annotations (despite improved performance). Yet, metacognitive monitoring includes a variety of additional aspects as well (e.g., strategy selection, self-evaluations of learning, Afflerbach & Cho, 2009). Given the promising results found here, additional metacognitive prompts should be used in future work.

Last but not least, a substantive limitation stemmed from our efforts to analytically engender strategy use, by asking students to engage in relevance-focused processing or evaluation or intertextual integration or metacognitive monitoring, when in reality all of these processes are used throughout multiple text learning (Anmarkrud et al., 2013; Wolfe & Goldman, 2005). Although this was done to allow for a more focused and dedicated examination of the manifestations of each strategy approach, this may have limited students' processing (although students were still able to engage any strategy during multiple text use, regardless of task condition). Different means of instantiating strategy use should be considered in future work.

5. Conclusions

This study examined how two scaffolds – providing students with a (a) highlighting and annotation tool and (b) task instructions to engage in particular modes of strategic processing – could foster learning from multiple texts, an essential 21st century skill. Instructing students to engage in metacognitive monitoring resulted in fewer highlights and annotations produced; while instructions to evaluate information during reading, indeed, resulted in students producing more evaluative annotations and accessing more references during reading. Although no difference in multiple text comprehension and integration performance were found across strategy conditions, models found students' highlighting of information necessary to aid evaluation (i.e., source information, statistical evidence) and their accessing of references during reading to predict multiple text comprehension and integration outcomes. Thus, this study contributes to the literature on learning from multiple texts by demonstrating the promise of highlights and annotations in facilitating such learning. It also contributes to the literature on digital annotations by suggesting that these may be improved based on the task instructions introduced. From a theoretical standpoint, this study provides support for the importance of metacognitive monitoring and behavioral engagement as processes identified in prior work and supporting learning from multiple texts. From a design stand point, this study suggests the need to build multiple text annotation platforms, with scaffolds supporting evaluation and metacognition built in (Barzilai et al., 2021).

Credit author statement

Alexandra List: Conceptualization; study and materials design; data collection, coding, and Formal analysis; writing, Chang-Jen Lin: data coding and analysis; writing support.

Data availability

Data will be made available on request.

Appendix A

Excerpts from Study Texts

Understanding the History of Mass Incarceration

Dr. John Williams, Professor of American History.
Princeton University.

Feature Article in the Economist.

According to Bureau of Justice statistics, in 2018 there were over 2.3 million people in U.S. prisons. This corresponds to 698 people being incarcerated out of every 100,000 citizens, reflecting 1 out of every 143 U.S. adults.

Much of the growth in the prison population has occurred over the last 40 years, with those incarcerated for drug-related crimes increasing from 40,000 in 1980 to over 400,000 in 2017. This increase is attributable to the U.S. government's War on Drugs, which created mandatory minimums for drug-related crimes

U.S. is Unique in the World in Terms of Mass Incarceration

Dr. Sam Campbell, Professor of Criminology.
Dartmouth University.

Feature article in Foreign Policy Review.

More people are incarcerated in the U.S. than anywhere else in the world. Analysis from the United Nations Office on Drugs and Crime (2019), found that the U.S. has less than 5% of the world's population, but almost 25% of the world's prisoners.

In part, this is due to the "tough on crime" culture of criminal justice in the U.S., wherein individuals receive significantly longer sentences than elsewhere in the world. On average, individuals convicted of committing fraud serve 10.9 months in jail in England, 30 months in Australia, and 44 months in the U.S.

Facts and Myths about Mass Incarceration in the United States

Dr. Mark Miller, Professor of Public Policy.
University of Pennsylvania.

Feature article in the New Yorker.

Among the most popular reforms proposed for mass incarceration has been to pardon those convicted of drug-related crimes. But, the unfortunate reality is that those incarcerated for drug-related crimes represent only a minority, or 20%, of the prison population (U. S. Department of Justice, 2017). Moreover, only about half of these individuals, or 12% of U.S. prisoners, are non-violent drug offenders, meaning that such a policy would be insufficient to address the problem of mass incarceration.

Rather than pardoning individual offenders, reformers have increasingly argued for understanding mass incarceration as a cultural problem with the structure of our criminal justice system

The Injustice of Criminal Justice: Mass Incarceration in the United States

Dr. Aaron Lewis, Professor of Sociology.
Cornell University.

Feature Article in the Atlantic Magazine.

The mass incarceration crisis in the U.S. includes not only the size of the prison population, but also its demographic makeup. Although Black and Latino people make up only 32% of the US population, they comprised 56% of all incarcerated people in 2015. This disproportionately comes from **over-policing**, or intense surveillance of minoritized communities by police. For example, while Black and White people use drugs at approximately the same rate, Black people constitute 36% of drug arrests and 46% of drug-related convictions, despite only constituting 12% of the population (Tsai, Wu, Bråten, & Wang, 2022).

The punishment-oriented culture of the U.S. also contributes to mass incarceration, wherein we incarcerate individuals for even minor crimes ...

Appendix B

Screenshot of Highlighting and Annotation Interface.

Please read this text to write a research report about mass incarceration in the United States. Please be sure to highlight and comment on any **important information** in this text.

U.S. is Unique in the World in terms of Mass Incarceration

Dr. Sam Campbell, Professor of Criminology
Dartmouth University
Feature article in Foreign Policy Review

More people are incarcerated in the U.S. than anywhere else in the world. Country-by-country analyses found that the U.S. has less than 5% of the world's population, but almost 25% of the world's prisoners [1].

In part, this is due to the "tough on crime" culture of criminal justice in the U.S., wherein individuals receive significantly longer sentences than elsewhere in the world. On average, individuals convicted of committing fraud serve 10.9 months in jail in England, 30 months in Australia, and 44 months in the U.S.. Likewise, assault charges carry with them, on average, a 15.2 month sentence in England, a 20 month sentence in Australia, and a 60 month sentence in the U.S. [2]. These differences in sentence length compound, leaving the U.S. with many more prisoners than elsewhere in the world.

Parole, or community supervision for low risk offenders, is a key component of the U.S.'s large prison population. Although seemingly a compassionate alternative to incarceration, parole extends individual responsibility for crime beyond the prison walls. Because parole extends individual responsibility, those on parole are often sent back to prison for minor violations. A 2016 study found that 4 out of 10 people incarcerated in the U.S. were on parole at the time of arrest, many for only technical violations [3].

Study from the Pew Charitable Trusts. Pew Charitable Trusts is an independent, non-profit, non-governmental organization. Its stated mission is to serve the public interest by improving public policy, informing the public, and invigorating civic life.



This is a sample comment.

This is another comment

Note: Yellow and green highlighting options were provided so that our digital annotation tool better mirrored those commercially available (e.g., within e-textbooks). The total number of yellow and green annotations that students rendered was collapsed in analyses.

Appendix C

Excerpts from Student Responses

Excerpts from student response:

Participant 139:

"Mass incarceration in the United States has been shown to be a topic of controversy. While many people propose solutions to lowering the number of people incarcerated, such as pardoning those convicted of drug-related crimes [Text 3] or the push to have more criminals sent on parole or probation [Text 1], many problems arise. In reality, drug offenders make up only a small percentage of the prison population (from the first article) [Text 3], and statistics have shown (according to the fourth article) that 4 out of 10 people incarcerated in a certain year were on parole at the time of arrest [Text 2]. However, parole and probation are much cheaper for the government (third article) [Text 1], so these ideas seem more pleasing at a glance ..."

Participant 202:

"The United States has the largest prison population in the world, with over 2.3 million people currently incarcerated. [Text 1] Although the U.S. holds only 5% of the global population, we have 25% of the global prison population. [Text 2] This disproportionality is the result of centuries of racist, anti-Black violence that criminalizes Black, Indigenous, and Latinx people. [Text 4]

Mandatory minimums, introduced during the 'War on Drugs,' set minimum sentencing limits, which results in courts sending people to prison for significant periods of time, even for minor offenses. [Text 1] ... Even though white people and Black people use drugs at the same rates, Black people are significantly more likely to be arrested for the same crime, demonstrating the intentional racism of the carceral system. The rhetoric of 'tough on crime' and 'law and order' refers to racist overpolicing and hyperincarceration of Black people and other marginalized populations [Text 4] ..."

Note: Sources of origin (e.g., [Text 1]) were designated by the researcher. Language indicating inter-textual linking is italicized and underlined.

Appendix D

Exploratory Analyses

In addition to examining which specific information students highlighted and which types of annotations they rendered, we were interested in considering whether the total number of highlights and annotations that students produced predicted performance. These analyses were guided by recent work on the importance of behavioral engagement when learning from multiple texts (Bråten et al., 2014; List et al., 2019).

Here we include four regression models predicting each outcome measure of interest (i.e., students' research report comprehensiveness and degree of integration and objective comprehension and integration performance). For each model, strategy condition was still controlled for, with membership in the intertextual linking, evaluation, and metacognitive monitoring strategy conditions, relative to the relevance processing condition, indicator coded and entered in Step 1. Then, the total number of highlights and annotations that students produced was entered in Step 2.

Research Report Comprehensiveness

The model predicting the number of arguments reflected in students' research reports was significant, $F(5, 230) = 6.14, p < 0.001, R^2_{adj} = 0.10$. The total number of highlights that students produced ($\beta = 0.31, p < 0.001$) was the only individually significant predictor in the model. See Table D1 for a model summary.

Table D.1
Model Predicting Number of Arguments Based on Quantitative Indicators

	<i>B</i>	<i>SE(B)</i>	β	Sig.
InterText Condition	0.36	0.39	0.07	0.35
Evaluation Condition	-0.22	0.39	-0.04	0.57
MetaCog Condition	0.74	0.40	0.14	0.07
Total Highlights	0.07	0.02	0.31	<0.001***
Total Annotations	0.04	0.03	0.09	0.18

Research Report Integration

The model predicting the volume of integration included in students' writing was not significant, $F(5, 230) = 1.91, p = 0.09$.

Objective Comprehension Performance

The model predicting objective comprehension performance was significant, $F(5, 243) = 6.16, p < 0.001, R^2_{adj} = 0.09$. Individually significant predictors in the model included students' membership in the intertextual linking task condition ($\beta = -0.16, p < 0.05$), with this associated with decreased performance, relative to membership in the relevance processing condition, as well as the total number of highlights ($\beta = 0.22, p < 0.01$) and annotations ($\beta = 0.15, p < 0.05$) rendered. See Table D2 for a model summary.

Table D.2
Model Predicting Objective Comprehension Performance Based on Quantitative Indicators

	<i>B</i>	<i>SE(B)</i>	β	Sig.
InterText Condition	-0.96	0.48	-0.15	0.04*
Evaluation Condition	-0.66	0.48	-0.10	0.17
MetaCog Condition	0.97	0.50	0.15	0.06
Total Highlights	0.06	0.02	0.22	0.003**
Total Annotations	0.07	0.03	0.15	0.03*

Objective Integration Performance

The model predicting total integration scores was also significant, $F(5, 243) = 6.17, p < 0.001, R^2_{adj} = 0.09$. Individually significant predictors included the total number of highlights ($\beta = 0.25, p < 0.001$) and annotations ($\beta = 0.16, p < 0.05$) that students generated. See Table D3 for a model summary.

Table D.3
Model Predicting Objective Integration Performance Based on Quantitative Indicators

	<i>B</i>	SE(<i>B</i>)	β	Sig.
InterText Condition	−0.48	0.51	−0.07	0.35
Evaluation Condition	−0.27	0.51	−0.04	0.60
MetaCog Condition	0.70	0.53	0.10	0.19
Total Highlights	0.07	0.02	0.25	<0.001***
Total Annotations	0.08	0.04	0.16	0.02*

Discussion

Exploratory analyses examining just the quantity of students' highlights and annotations found these to predict the number of arguments that students included in writing as well as objective comprehension and integration performance. These results seem to provide evidence for the construct of *behavioral engagement*, defined as students' "active, observable involvement in academic tasks ... typified by time, effort, persistence, and productivity" (Bråten et al., 2018; Bråten et al., 2022, p. 288). That is, beyond the cognitive or processing benefits offered by students' highlighting and annotating of texts, these, more broadly, may have reflected students' effort expenditure and willingness to complete the activities required to meet task demands, additionally offering an observable means of analyzing students' engagement.

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